CD61203 Project

Essential Tools for Scientific Computing: term project guidelines for students

Project Submission Rules

Every project must be maintained as a Git repository.

1. **Check the directory** at the start of the session: `rollnumber-project-name`, for example, `23XX12345-project1`. Make sure to keep all your files in this directory only.
2. **Commit regularly as you work.** After each meaningful step, and at least once every 10 minutes, save work in git, and use short, descriptive commit messages, such as “Add bibliography entries”. If you use Overleaf, download the project often, unzip it into the root of your repository (not as a `.zip` file), and commit (if compressed, uncompress the files before saving the project to git). A single commit at the end, or no incremental history, may result in a reduction in marks.
3. **Create a directory named `tmp-check`** in the root of your repository.
4. **Add files to `tmp-check` when instructed** during the session. These files are used to verify progress during the session. Commit them immediately after adding them. Do not rename or modify them after the instructed time.

Note: Sharing the problem or solution with anyone is academic misconduct and will result in zero marks.

Project statement:

You are given a GitHub repository: https://github.com/huggingface/diffusers

Check the menu option Insights, and look at the plots for,

1. Contributors (commits over time per contributor)
2. Commits (commits per week)
3. Code frequency (weekly additions and deletions)
4. Top committers (commits per author)

Reproduce these four graphs using Python (matplotlib). Your figures must match the reference in chart type, axes, labels, series and values, to within a small tolerance. Exact pixel identity is not required.

Write a LaTeX document with the four figures, captions, and labels.

Deliverables (in your Git repository):

1. One subdirectory per figure (for example, `fig1-contributors/`). Each contains the raw data (CSV), the Python code, the generated image, and the matching reference image from the GitHub website.
2. `main.tex`: the four figures with captions and labels, each shown side by side with its reference figure.
3. `main.pdf`: compiles without errors.